

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28				

ARRAYS

Max Subarray Sum -

```

for (i to arr.length - 1)
  for (j = i to arr.length)
    curSum = 0
    for (k = i to k = j)
      curSum += arr[k];

```

```

    (maxSum < curSum)
    maxSum = curSum;

```

TC $\rightarrow O(n^3)$.

Dry Run -

arr = {-2, 1}

i = 0

j = 0

curSum = 0

k = 0

CS = 0 + (-2)
= -2

maxSum = -2

k = 1 X

j = 1

curSum = 0

k = 0

CS = -2

maxS = -2

k = 1

CS = -2 + (1) = -1

maxSum = -1

j = 2 X

j = 2, j > arr.length

k = 1

CS = 1

maxS = 1

2017 • Dictation

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

2] Prefix Array -

* Skip 3rd loop -

① prefix array calculate -

arr	2	-4	6	8	-10
prefix	2	-2	+4	+12	2

② $prefix[0] = arr[0]$ ③ $prefix[i] = prefix[i-1] + arr[i]$

④ Skipped of 3rd loop ->

cursum = 0 $i=0$? $prefix[i] : prefix[i] - prefix[i-1]$

Dry Run -

 $i=0$ $j=1$ $i=2$ $i=3$ $i=4$ $j=0$ $j=1$ $j=2$ $j=3$ $j=4$ $CS=2$ $CS=-4$ $CS=2$ $CS=8$ $CS=24$ $j=-1$ $j=2$ $j=3$ $j=4$ $j=-2$ $j=0$ $j=3$ $j=4$ $j=10$ $j=14$ $j=3$ $j=4$ $j=10$ $j=14$ $j=0$ $j=4$ $j=10$ $j=14$ $j=0$ $j=14$

2018 • JANUARY • VIKRANT

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

3] Kadane MaxSubArray Sum -

① We'll find cursum & maxsum in single loop only.

② If any value of subarray is -ve => replace with 0.

arr	-2	-3	4	-1	-2	1	5	-3
cursum	0	0	4	3	1	2	7	4
maxsum	0	0	4	4	4	4	7	7

LINKEDLIST

① Reverse -

Node prev = null

Node cur = head

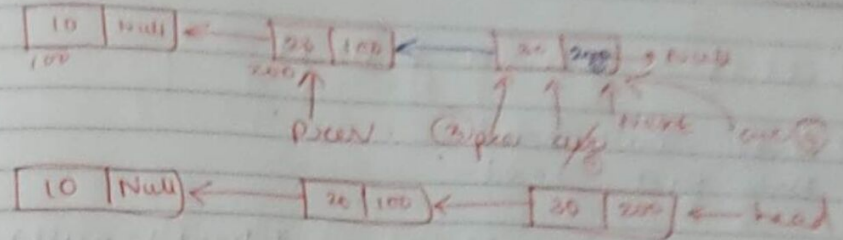
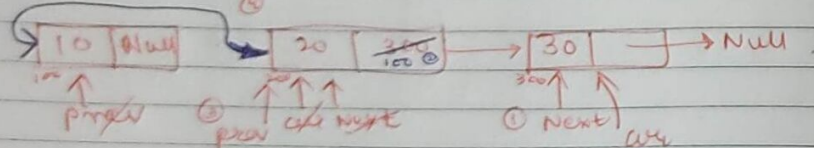
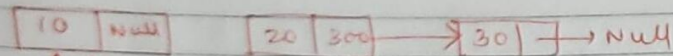
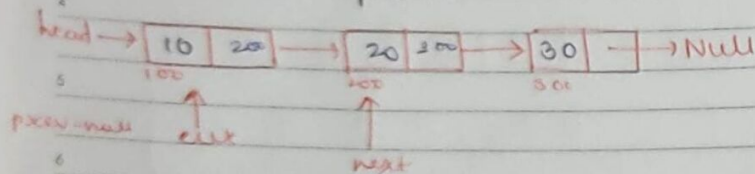
Node next;

```

while (cur != null) {
    cur.next = prev;
    prev = cur;
    cur = next;
}

```

head = prev



② Is Palindrome -

1. Find MidNode
2. Reverse 2nd half
3. Check 1st == 2nd

// 1.

Node slow = head;

Node fast = head;

```

while (fast != null && fast.next != null) {
    slow = slow.next;
    fast = fast.next;
}

```

Node mid = slow;

// 2. Reverse code just

Node cur = mid;

Node right = cur;

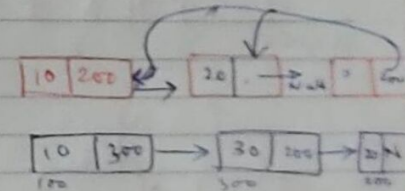
Node left = head;

// 3.

```

while (right != null) {
    if (right.data != left.data) {
        return false;
    }
    return true;
}

```



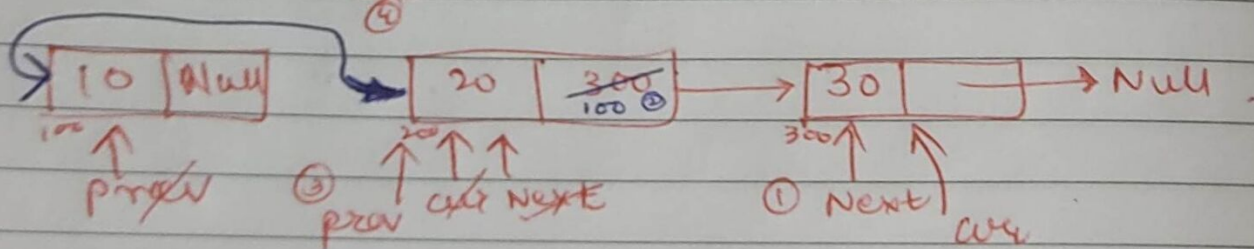
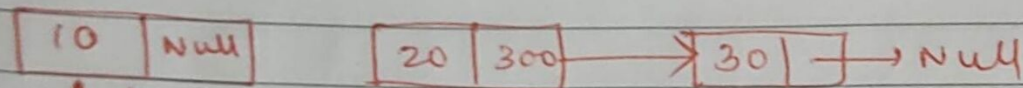
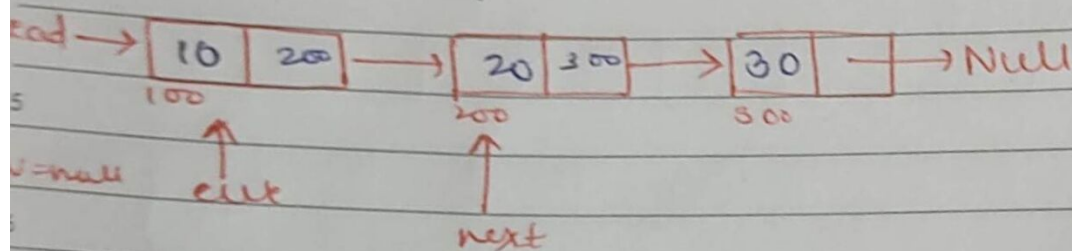
LINKEDLIST

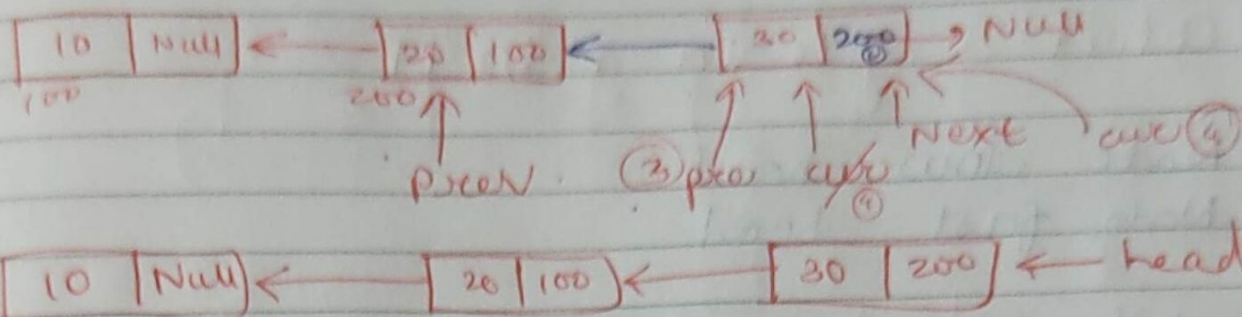
① Reverse

Node prev = null
 Node cur = head
 Node next;

while (cur != null) {
 ① cur.next = cur.next;
 ② cur.next = prev
 ③ prev = cur
 ④ cur = next.
 }

head = prev





#2 IS Palindrome -

1. Find MidNode
2. Reverse 2nd half.
3. Check 1st == 2nd.

// 1.

Node slow = head;

Node fast = head;

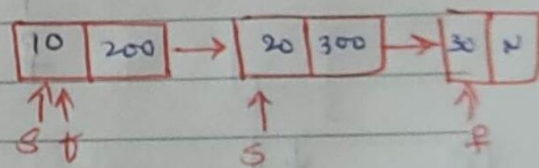
while (fast != null && fast.next != null) {

slow = slow.next

fast = fast.next

}

Node mid = slow;

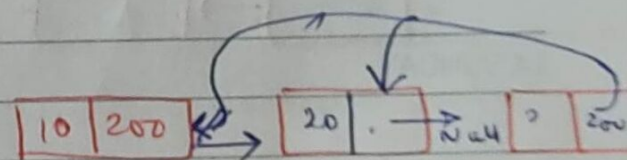


// 2. Reverse code just

Node cur = mid

Node right = cur

Node left = head



// 3.

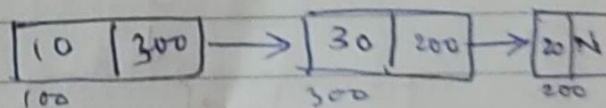
while (right != null) {

if (right.data != left.data) {

return false;

}

return true;



M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

③ Detect a cycle in a ~~list~~ LL -

Node slow = head

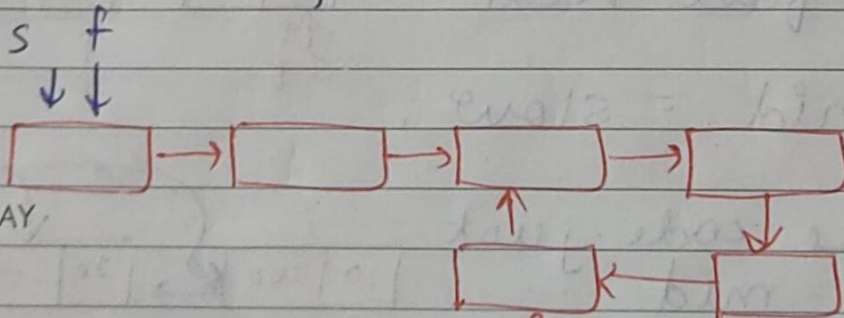
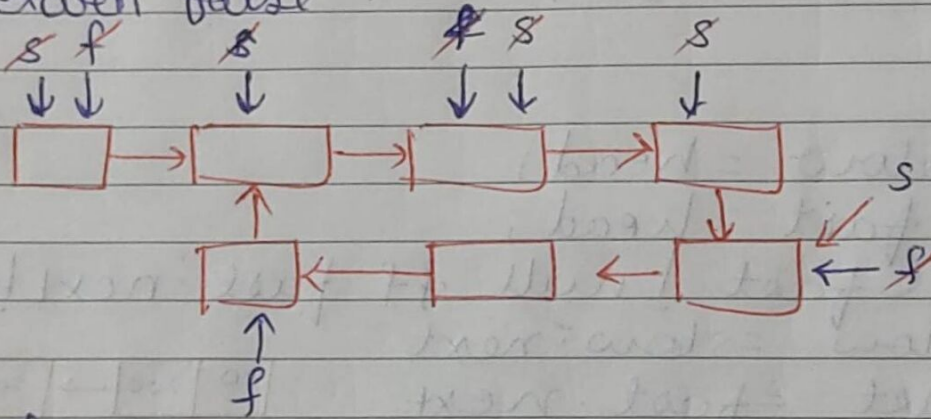
Node fast = head

```

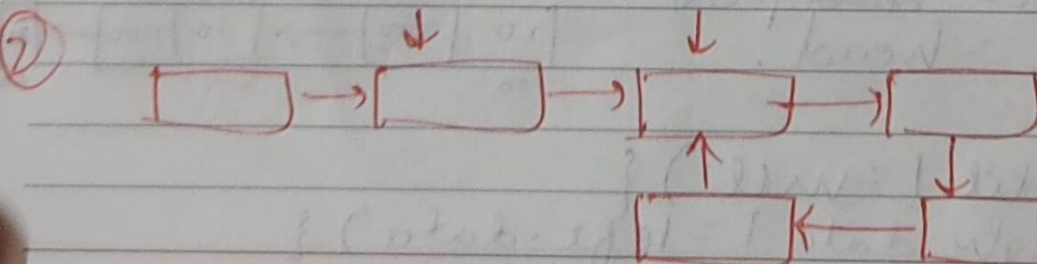
while (fast != null && fast.next != null) {
    slow = slow.next // + 1
    fast = fast.next.next // + 2
    if (slow == fast)
        return true
}

```

return false



14 SUNDAY

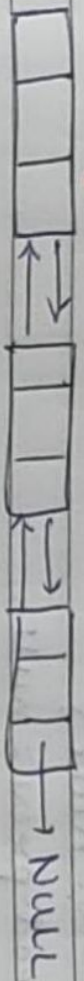


M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28				

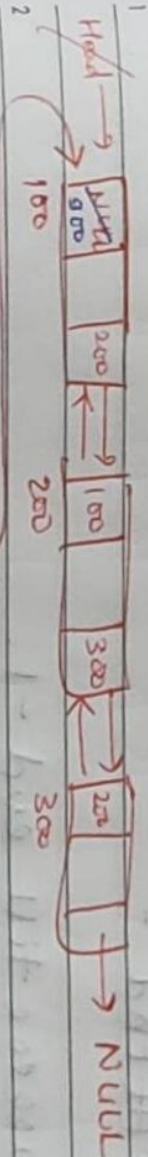
prev.next = null;

No → false.

#5 Doubly linked list - head, prev, next



① Insert At Start -



N.next = head = 100

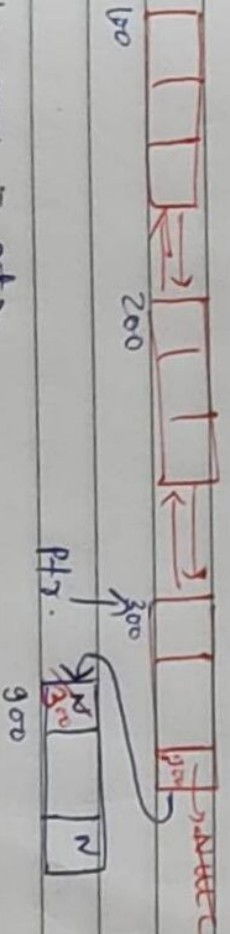
head.prev = N = 900

head = prev N.

Head

② Insert At End -

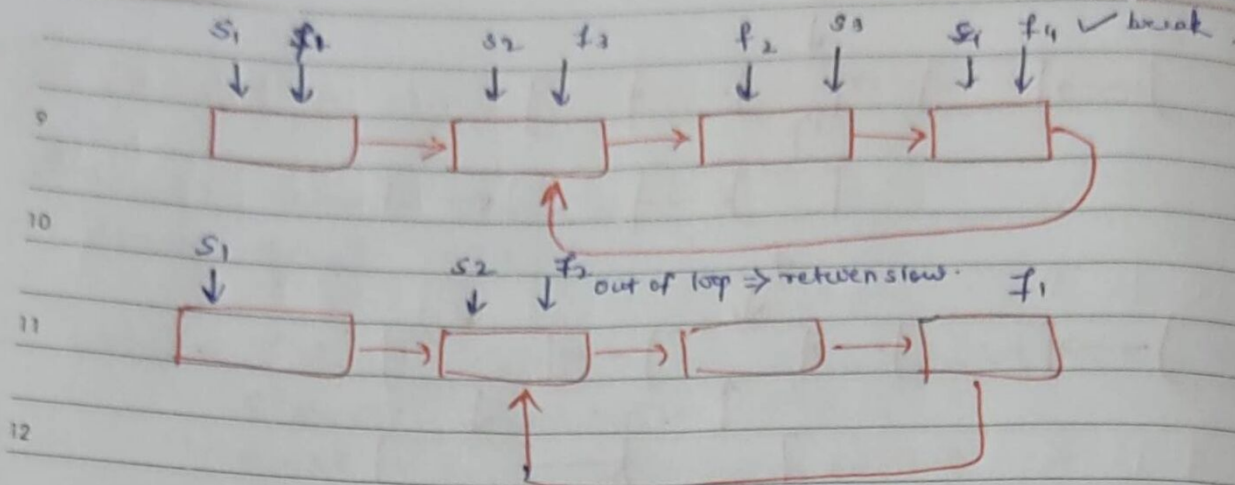
a. Traverse till end.



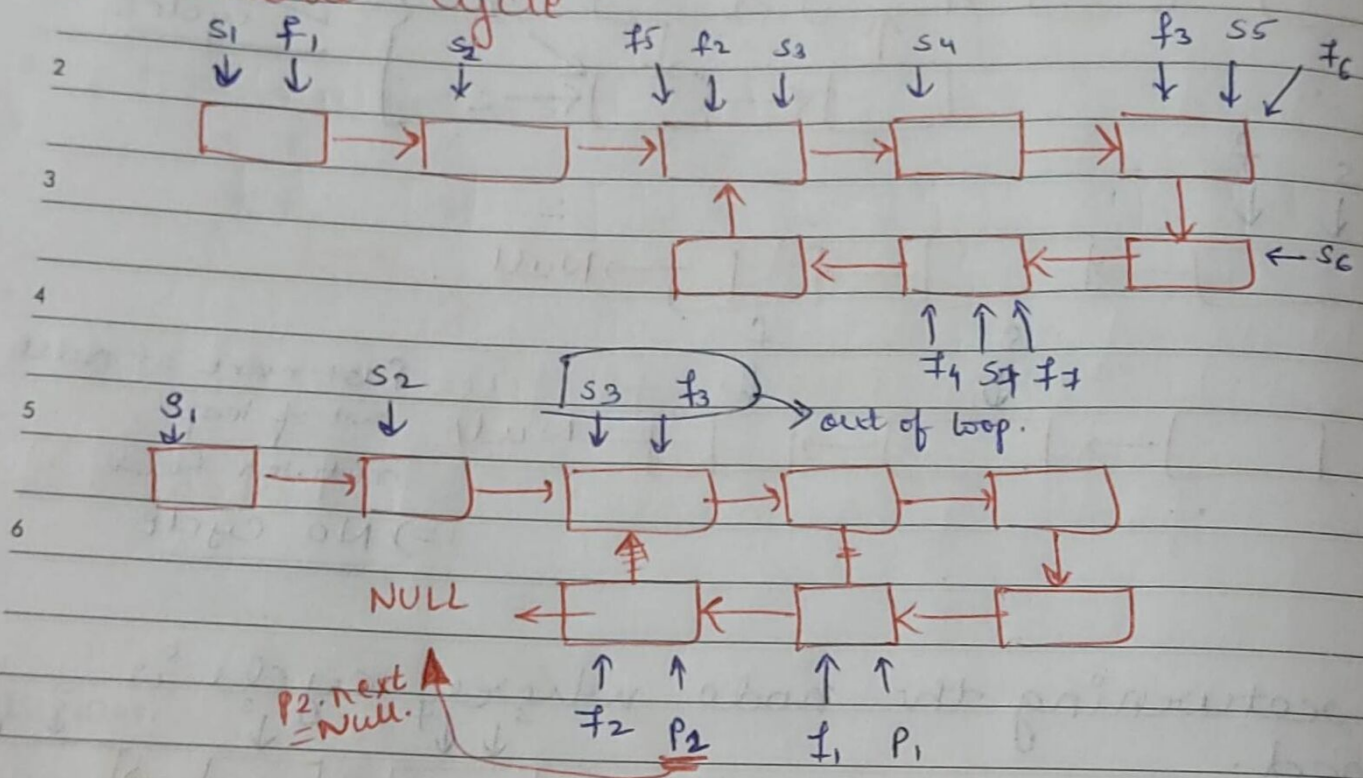
N.prev = ptr

ptr.next = N

N.next = null.



④ Remove Cycle -



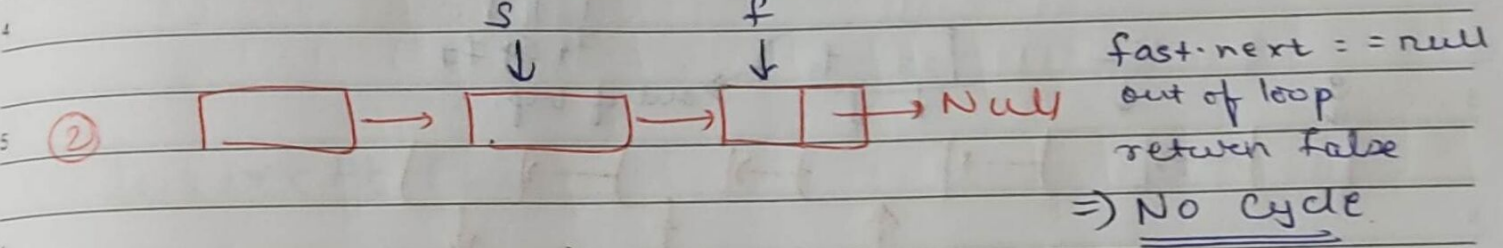
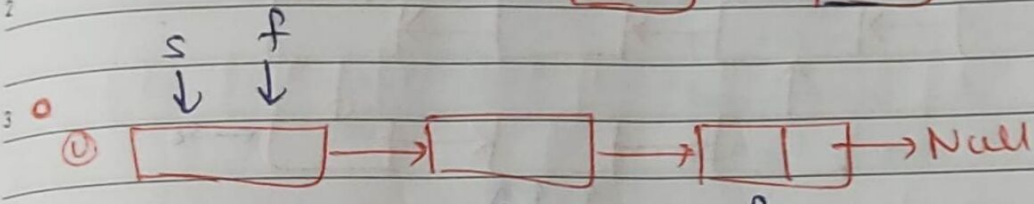
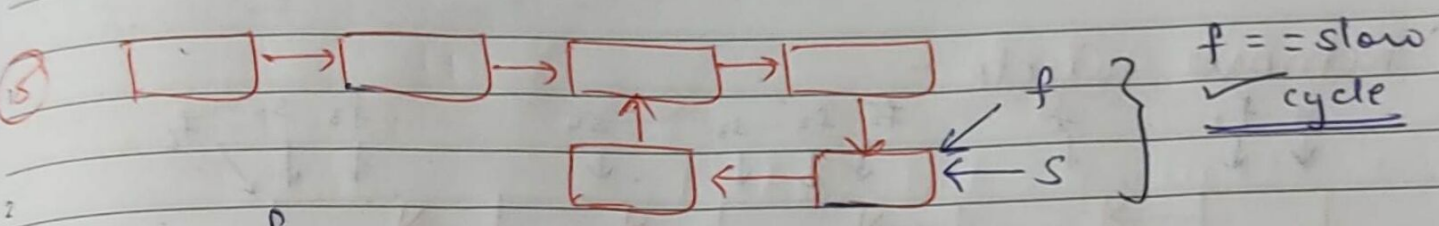
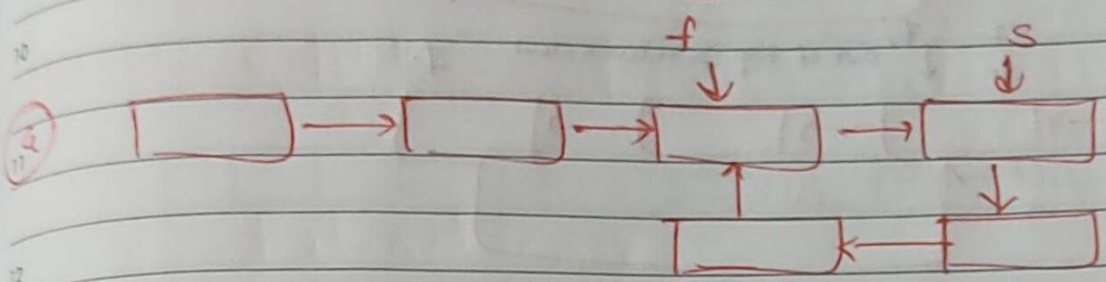
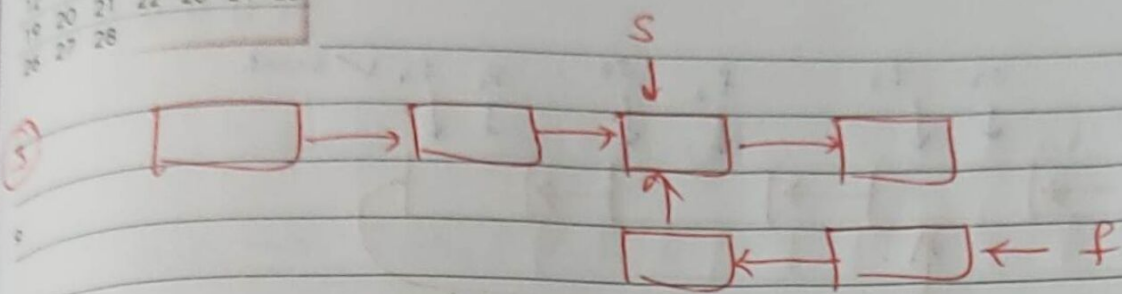
// 1. Detect cycle.

2. If it exists →

Yes → slow = head, prev = fast
while (slow != fast)

slow ++
prev = fast
fast ++

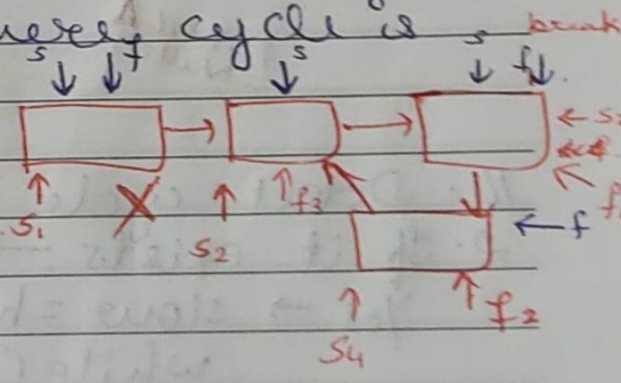
}



• for returning the node where cycle is formed.

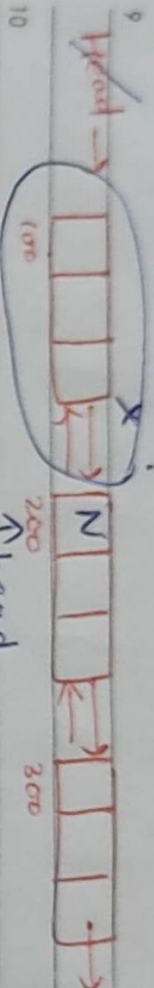
```

if (fast == slow) break;
slow = head
while (slow != fast) {
    slow = slow.next
    fast = fast.next
}
return slow;
  
```



M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

⑥ Delete At Start .



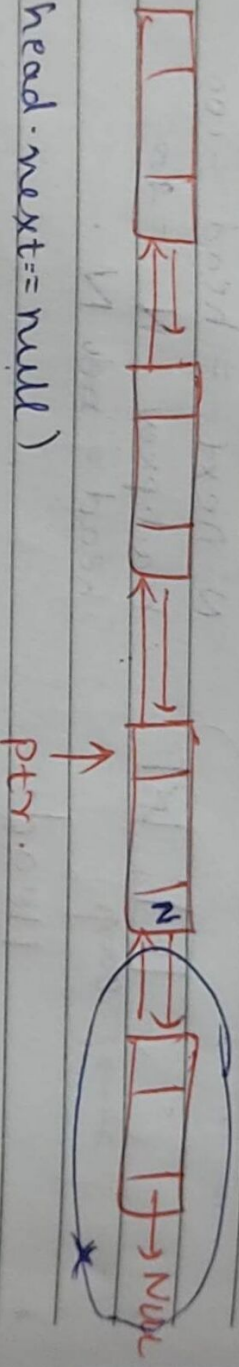
Head = head.next .

if (head == null)

head = null .

⑦ Delete At End -

1. Traverse till end - 1 .



if (head.next == null)

head = null

else

ptr.next = null

July 2019							August 2019						
01	02	03	04	05	06	07	08	09	10	11	12	13	14
15	16	17	18	19	20	21	22	23	24	25	26	27	28
29	30	31											

Wk-23 • Day (159-206)

Recursion

- 1) Base Case
- 2) Work
- 3) Inner calls.

* Print N Numbers -

$n = 10$.

- 1) Base Case $\rightarrow (n == 0)$ return;
- 2) print no. $\rightarrow \text{sop}(n)$
- 3) Inner calls $\rightarrow \text{print}(n-1)$

10 9 8 1

print(10)

10
↘ print(9)

9
↘ print(8)

8
↘ ...

print(1)

1
↘ print(0)
return

Sunday 09

JUNE

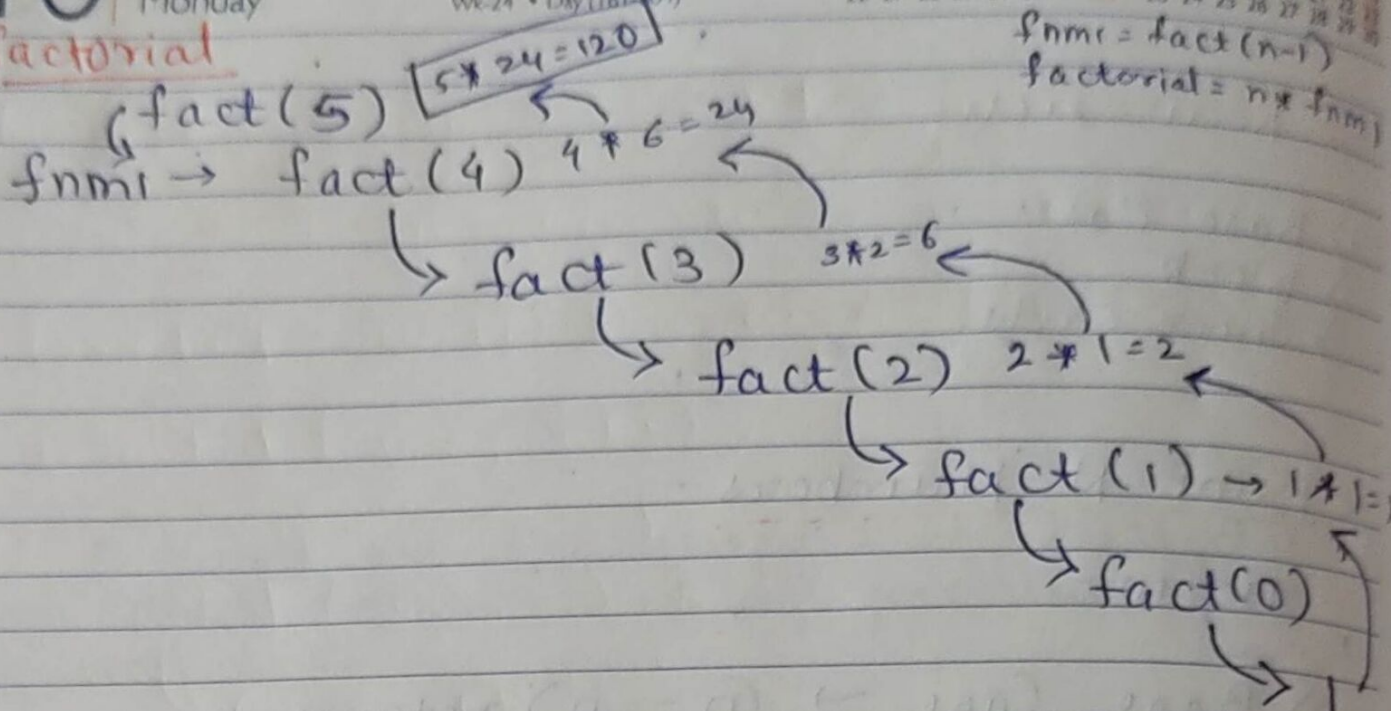
10 2019
Monday

Wk-24 • Day (161-104)

05	May 2019	06	June 2019
Wk M T W T F S S	Wk M T W T F S S	Wk M T W T F S S	Wk M T W T F S S
18 19 20 21 22 23 24	22 23 24 25 26 27 28	29 30 31 1 2 3 4	5 6 7 8 9 10 11
25 26 27 28 29 30 31			

factorial

7am



$fact(n) = fact(n-1)$
 $factorial = n \times fact(n-1)$

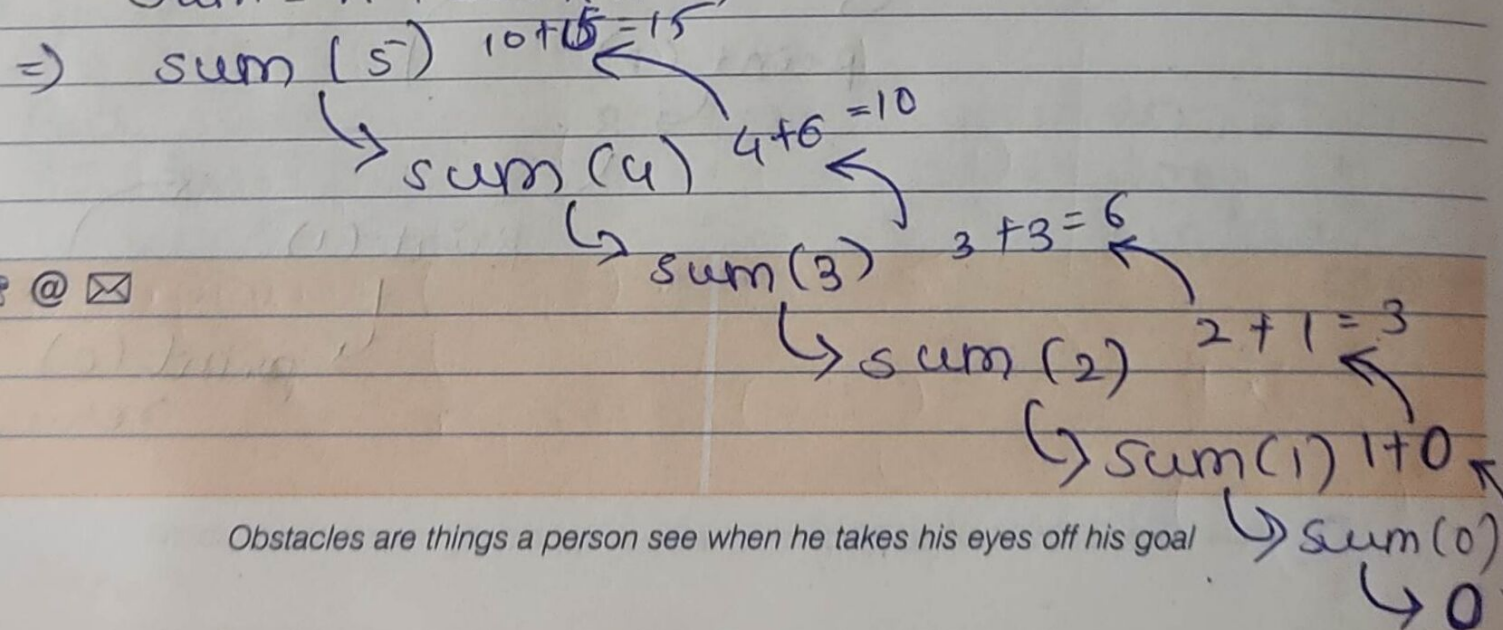
Sum of Natural Numbers -

sum(5)

TC = $O(n)$
 SC = $O(n)$

if (n == 0)
return 0;

$\Rightarrow sum(n-1)$
 $sum = n + sum(n-1)$



Obstacles are things a person see when he takes his eyes off his goal

July 2019							August 2019							
	T	W	T	F	S	S	W	M	T	W	T	F	S	S
1	2	3	4	5	6	7	31				1	2	3	4
8	9	10	11	12	13	14	32	5	6	7	8	9	10	11
15	16	17	18	19	20	21	33	12	13	14	15	16	17	18
22	23	24	25	26	27	28	34	19	20	21	22	23	24	25
29	30	31					35	26	27	28	29	30	31	

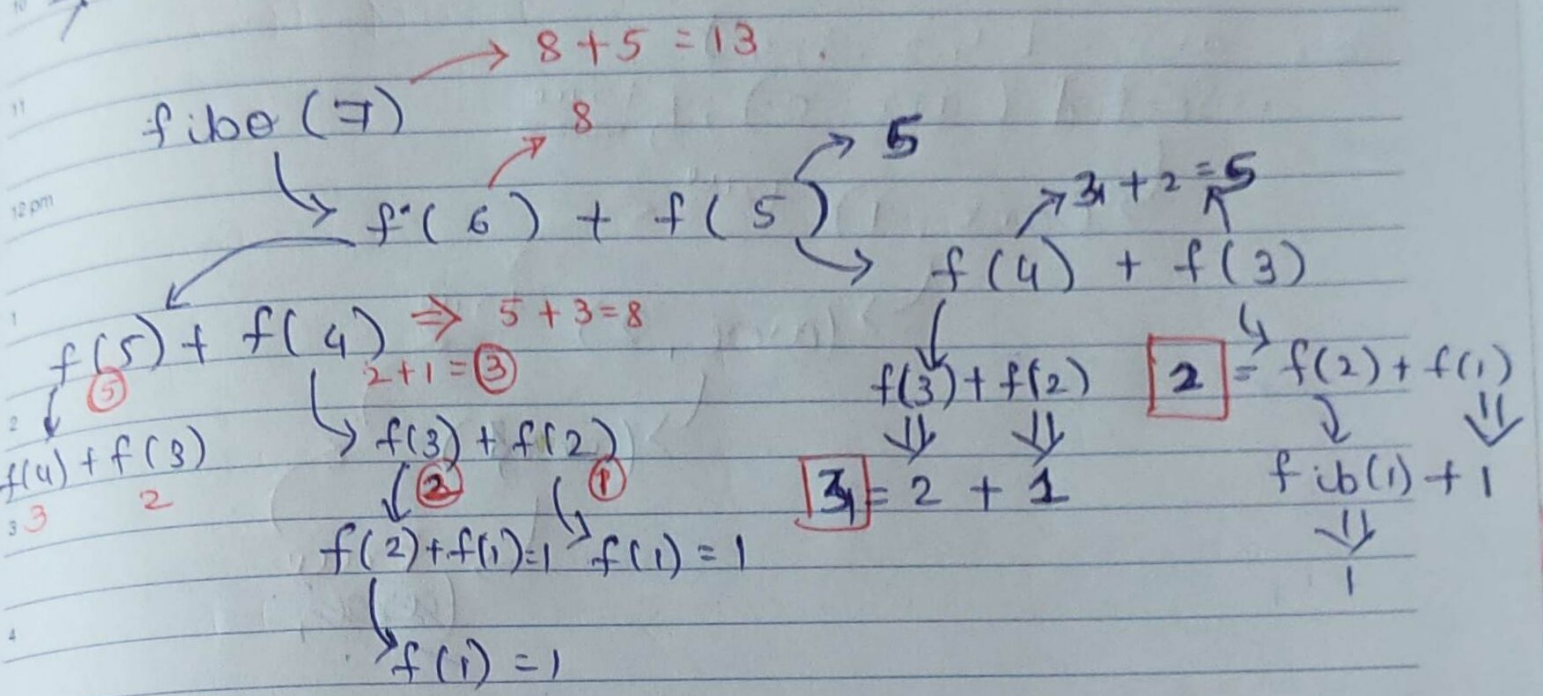
Wk-24 + Day (162-203)

JUNE
2019
Tuesday

Fibonacci Series

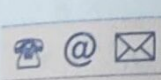
1) $\{n == 0\}$ return 0;
 $\{n == 1\}$ return 1;

2) $fib = fib(n-1) + fib(n-2)$



No. of calls = $2^n = 2^5 = \underline{\underline{32}}$ (fib(5))

TC - $O(2^n)$



JUNE
12 2019
Wednesday

Wk-24 • Day (163-202)

May 2019							June 2019						
Wk	M	T	W	T	F	S	S	Wk	M	T	W	T	F
18			1	2	3	4	5	22					
19	6	7	8	9	10	11	12	23	3	4	5	6	7
20	13	14	15	16	17	18	19	24	10	11	12	13	14
21	20	21	22	23	24	25	26	25	17	18	19	20	21
22	27	28	29	30	31			26	24	25	26	27	28

Check array is Sorted -

1) $(i == \text{arr.length} - 1)$ return ~~false~~ true

2) $(\text{arr}[i] > \text{arr}[i+1])$ return false

3) $\text{isSorted}(\text{arr}, i+1)$

So, $\text{arr} = [1, 2, 3, 4]$

$\text{isSorted}(\text{arr}, 0)$; true

↳ $(\text{arr}, 1)$ true

↳ $(\text{arr}, 2)$ true

↳ $(\text{arr}, 3)$

↳ true
~~(arr, 4)~~

Power x^n -

TC = $O(n)$

$(\text{base} == 0)$ return ~~base~~ 1

return $\text{power}(\text{base}, \text{exp} - 1)$

@ power(2, 3); $2 * 4 = 8$

base * $\text{power}(2, 2)$ $2 * 2 = 4$

↳ $\text{power}(2, 1)$

↳ 2

TC = $O(n \log n)$
if exp = even
↳ $x^{n/2} * x^{n/2}$
odd $\Rightarrow x * x^{n/2} * x^{n/2}$

A dream with courage is innovation... A dream without courage is delusion

July 2019

08

August 2019

JUNE

2019

13

Thursday

Wk-24 • Day (164/201)

Tiling Problem -

($n == 0$ || $n == 1$) return 1;

// Vertical arrangement

int fnm1 = tilingProb (n-1)

// Horizontal

int fnm2 = tilingProb (n-2)

return fnm1 + fnm2;

So,

tilingProb (4) = 5

$tP(3) + tP(2)$

$tP(2) + tP(1)$

$tP(1) + tP(0)$

$tP(1) + tP(0)$

1 + 1 = 2

1 + 1 = 2